

Anirudh Suresh Kumar

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EDUCATION

Indraprastha Institute of Information Technology, Delhi (IIITD)

06/25

B.Tech in Computer Science with Artificial Intelligence

GPA: 8.76/10

EXPERIENCE

Software Engineer | *TBO.com*

06/25 – Present

- Automated customer support query processing using LLMs to analyze and resolve email inquiries, reducing resolution time and operational costs while eliminating manual triage
- Engineered a real-time notification system for legacy tickets using Redis Streams and Golang; developed and integrated a dynamic React UI to deliver a seamless user experience.
- Developed a standardized metrics pipeline with automated alerting by building a reusable library that collects, processes, and visualizes operational data through Kafka, Timestream, and Grafana

Backend Intern | *Ripik.Ai*

05/24 – 07/24

- Detected an unsanitized input vulnerability in an API endpoint allowing lateral privilege escalation
- Remediated four critical AWS misconfigurations, hardening S3 bucket security, preventing malicious activities and protecting administrative access
- Produced technical documentation highlighting SQL injection risks in existing code and recommended secure coding practices for mitigation

Undergraduate Researcher | *Network and Security Lab, IIITD*

05/23 – 06/25

- Designed and implemented ViNET, a covert communication system using Video-over-LTE to bypass internet shutdowns.
- Achieved 200 kb/s throughput, outperforming over 300× faster than prior audio-based systems like Dolphin's 600 b/s
- Ensured end-to-end encryption and resistance to machine learning-based traffic analysis, validating robustness against state-of-the-art ML classifiers
- Peer-reviewed paper on the same set to appear in IEEE Euro S&P 2026

PROJECTS

Distributed Systems | *Python, Docker, gRPC*

01/24 – 04/24

- Implemented Raft Algorithm for creating a distributed key-value database
- Stabilized leader election using leader lease for linearizable reads
- Recreated real-time testing setup using docker compose, and later reproduced using VMs on GCP
- Created a Low-Level Low-Latency Group Messaging Application using ZeroMQ
- Implemented a scalable MapReduce-based KMeans algorithm with dynamic worker allocation for efficient parallel processing

rm - temporary rm | *Rust, Clap, Serde, Unix*

01/25 – 02/25

- Created a Rust CLI tool for reversible file deletion by relocating files to a temporary directory
- Used Serde for structured history logging, time-based file recovery, and purge operations
- Handled symlinks, directory structures, and permission-related edge cases across Linux file systems

Parallel Runtimes | *C/C++, HCLib, Argobots*

01/24 – 04/24

- Implemented 4 different implementations of async-await paradigm, optimized for performance and energy efficiency
- Made a custom dynamic queue-rebalancing runtime using Argobots as backend
- Achieved speedup over baseline by as much as 30%

Fluid Simulation | *C++, OpenGL, Compute Shaders*

09/24 – 12/24

- Created a 2D Fluid Simulator from scratch using OpenGL by utilizing the PCISPH algorithm
- Implemented Dynamic addition to the simulation, along with a Compute Shaders based implementation
- Able to achieve more than 60fps at 25,000 particles in the simulation

SKILLS

Languages: Golang, C/C++, Rust, Java, SQL, Python, HTML/CSS, JavaScript/TypeScript, $\text{\LaTeX}$

Tools: PostgreSQL, Unix Shell, Linux, Git/GitHub, GitHub Actions, Vim, AWS, GCP, MySQL, Redis, MongoDB

Frameworks: Fiber, Svelte, SvelteKit, TailwindCSS, React, Selenium, FastAPI

Libraries: gRPC, Pandas, NumPy, Matplotlib, sklearn, boto3, LibGDX, clap, Pygame,

POSITIONS OF RESPONSIBILITY

Coordinator | *d4rk0de and OWASP IIITD Student Chapter*

03/23 – 05/24

- Cybersecurity Club of the College, Achieved AIR 6 on CTFtime for CTF(Capture the Flag) in 2023
- Head of the OWASP Foundation's Student chapter at IIITD